

Bachelor in Computer Applications

Course Title: **Advanced Java Programming**

Code No: Semester: **7**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Why do we need interface? Define an interface named shape with area () method. Create a class named Rectangle with length and breadth circle with radius that implements shape interface. Now create the object of both classes and display their area.
2. Write a Java program to create a UI with three labels and three text boxes for First Number, Second Number and Result. Also put Four buttons (Add, Subtract, Multiply, and Divide) in the form and handle the event accordingly.
3. What are the various tags used in writing JSP programs? How HTML forms can be processed using JSP? Explain with example.
4. Describe the chain of constructor with example.
5. Write a Java program to create a multi-threaded programs by extending Thread class. Create at least three threads.
6. What is layout manager? Explain working of flow layout with example.
7. Describe about exception handlings with example.
8. How prepared statements differs from statements? Explain use of prepared statements with example.
9. Discuss the steps of writing JavaFX programs.
10. Write a socket program using TCP where client sends an integer to server and server response whether it is positive or negative.
11. How distributed programs can be written in Java? Explain RMI architecture.
12. Write short notes on a. Final keyword b. Object class